

UNIT V

19. List any five favorite UML diagrams and state why it has drawn your attention.

Or

20. Draw a neat sketch to understand the e-trading system using Sequence diagrams.
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5633030

B.Tech. DEGREE EXAMINATION, MARCH 2021.

Third Semester

Computer Science Engineering

OBJECT ORIENTED PROGRAMMING AND DESIGN

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. List the advantages of object oriented programming.
 2. List down the features of object oriented programming.
 3. Write any two characteristics of a constructor.
 4. What is the use of destructor?
 5. Write any two situations that you represent for exception handling.
 6. What is the advantage of using function template?
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7. How reusability is carried off in C++?
8. Write down the behavior of virtual class.
9. What are the classes that help us handling files?
10. Draw the symbols that are used in activity diagrams to represent start and stop.

PART B — (5 × 11 = 55 marks)

Answer ALL questions.

All questions carry equal marks.

UNIT I

11. Compare and contrast object oriented programming language and procedure oriented programming language.

Or

12. Compare “for” and “while” statements. Illustrate their behavior through well written program/code and discuss their performances in a line or two.

UNIT II

13. Write the program to demonstrate the behavior of function overloading.

Or

14. Discuss the need for abstract class and illustrate the same using proper program.

UNIT III

15. Argue that dynamic memory allocation is your favorite choice of memory allocation mechanism. Discuss on the operators that does the job of dynamic memory allocation.

Or

16. Write code to demonstrate the accessibility of base and derived class entities using base class pointer.

UNIT IV

17. Tabulate the file access modes and explain how they behave.

Or

18. Differentiate error and exception. Write a code to handle the exception for user defined “Age Not Suitable for Voting”. Exception.

UNIT - V

19. Define patterns. Explain the different types of patterns. (11)

Or

20. Design a problem statement for Library management system. Draw the UML. Use Case, Class diagram and Activity diagram. (11)
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B.Tech. DEGREE EXAMINATION, AUGUST 2022.

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Computer Science and Engineering

OBJECT ORIENTED PROGRAMMING AND DESIGN

(2013 - 14 Regulations)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL the questions

1. What is an object? Give an example.
2. Write a C++ program that will take temperature in Fahrenheit and display it in Celsius.
3. Give any two operators that cannot be overloaded.
4. What is hybrid inheritance in C++?
5. State the characteristics of arrays.
6. Define virtual destructors. Write an example.
7. Mention the difference modes of file operations.

8. What is the main purpose of template in C++?
Give an example.
9. Highlight the use of state chart diagrams.
10. Specify the OOD goodness criteria.

PART B — ($5 \times 11 = 55$ marks)

Answer ALL questions, ONE question from each unit

UNIT – I

11. List the advantages and usage of Object Oriented Programming. (11)

Or

12. Enumerate the different looping statements in C++ with suitable examples. (11)

UNIT – II

13. Write the syntax of binary operator overloading. How many arguments are required in the definition of an overloaded binary operator by using member and friend function? Write a program to multiply two complex numbers using operator overloading. (11)

Or

14. Elaborate the different types of inheritance in C++ with suitable example. (11)

UNIT – III

15. Design a program in C++ that counts the frequency of words on an input stream and produce a list of (word, count) pairs in frequency order as output. (11)

Or

16. An election is contested by five candidates. The candidates are numbered 1 to 5 and the voting is done by marking the candidate number on the ballot paper. Write a program to read the ballots and count the votes cast for each candidate using an array variable count. In case, a number read is outside the range 1 to 5, the ballot should be considered as a 'spoilt ballot' and the program should also count the number of spoilt ballots. (11)

UNIT – IV

17. Write a function template for finding the largest number in a given array. The array parameter must be of generic data types. (11)

Or

18. Discuss about the exception handling mechanism in C++ with a program. (11)